Architecture

Raw evidence to analyst decisions, with explicit AI boundaries.

01 RAW SOURCES

customers (10,000) + loans (14,685) + bureau (22,979) + repayments (176,220) + collections (8,635). Immutable CSVs, SHA-256 checksums.

02 INGESTION & ACCOUNT FEATURES

pandas / PyArrow: typed validation, quality report, category normalization. Bureau aggregates by customer; repayment and collections by account. Left joins preserve 14,685 loans. Detailed sources retained for drill-down.

ANALYTICS

pandas: counts, default rates, observed exposure, cohorts and employment x income. Every rate retains its denominator.

DEFAULT MODEL

scikit-learn: logistic + histogram boosting. Customer-disjoint 60/20/20 split. Train-only preprocessing; AUC, KS, Gini; feature importance.

ANOMALIES

scikit-learn: peer rules + Isolation Forest. Unlabeled behavior, leave-one-out comparisons, deterministic review reasons.

03 STREAMLIT DASHBOARD / PLOTLY

Portfolio filters and risk/exposure charts | Account score, drivers and source histories | Ask the Portfolio | Anomaly review. Versioned parquet, joblib and JSON artifacts are verified before loading; startup never retrains.

SERVER-SIDE LLM / GROQ / GPT-OSS

HTTPS structured query planning. Send schema/metric definitions, categorical filter scope and analyst question only. Pydantic allowlists the result; pandas executes it; deterministic text cites exact local figures. No raw customer records or review notes leave the app.

SQLITE + CSV EXPORT

SQLAlchemy + sqlite3. Append-only, versioned confirm/dismiss events with UUID, note and UTC time. CSV download retains a review copy. Streamlit local files may reset; no durable cloud guarantee. Optional managed Postgres backend.

HOSTING, SECRETS & LIMITATIONS

Python 3.11 / Streamlit Community Cloud target. The LLM key stays server-side; SQLite needs no database credentials. Export review CSVs before a cloud restart/redeploy. Managed Postgres is optional for durable storage. Missing default dates, undated bureau snapshots and inconsistent collection chronology prevent prospective validation.